THE CONORMAL SHEAF OF A CLOSED SUBSCHEME COMPUTES THE FIRST EXT SHEAF OF ITS STRUCTURE SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0BQW (sources/stacks-duality/pinned-original-source.tex line 3620).

Allowed standard prerequisites (editorial).

- The second order thickening sequence. Same chapter, equation labelled equation-second-order-thickening, used at proof lines 3660-3666. Supplies the extension whose pushouts define the map.
- Sections of the derived Hom over an open and the interpretation of Ext^1 by extensions. Cohomology, Lemma labelled cohomology-lemma-section-RHom-over-U and Derived Categories, Lemma labelled derived-lemma-ext-1, cited at proof lines 3670-3674. Identify pushouts of the thickening with elements of the Ext sheaf.

Source reference. Original lemma 08DK: sources/stacks-chow/tags.txt, source lines 10640–10640. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 3644 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 06XV: sources/stacks-chow/tags.txt, source lines 8864–8864. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 3645 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let Y be a ringed space. Let $\mathcal{I} \subset \mathcal{O}_Y$ be a sheaf of ideals. Set $\mathcal{O}_X = \mathcal{O}_Y/\mathcal{I}$ and $\mathcal{N} = \mathcal{H}om_{\mathcal{O}_Y}(\mathcal{I}/\mathcal{I}^2, \mathcal{O}_X)$. There is a canonical isomorphism $c : \mathcal{N} \to \mathcal{E}xt^1_{\mathcal{O}_Y}(\mathcal{O}_X, \mathcal{O}_X)$.*

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Proof. Consider the canonical short exact sequence

$$0 \to \mathcal{I}/\mathcal{I}^2 \to \mathcal{O}_Y/\mathcal{I}^2 \to \mathcal{O}_X \to 0 \tag{0.1.1}$$

Let $U \subset X$ be open and let $s \in \mathcal{N}(U)$. Then we can pushout (0.1.1) via s to get an extension E_s of $\mathcal{O}_X|_U$ by $\mathcal{O}_X|_U$. This in turn defines a section $c(s)$ of $\mathcal{E}xt^1_{\mathcal{O}_Y}(\mathcal{O}_X, \mathcal{O}_X)$ over U. See Cohomology, Lemma 08DK and Derived Categories, Lemma 06XV. Conversely, given an extension

$$0 \to \mathcal{O}_X|_U \to \mathcal{E} \to \mathcal{O}_X|_U \to 0$$

of $\mathcal{O}_U$-modules, we can find an open covering $U = \bigcup U_i$ and sections $e_i \in \mathcal{E}(U_i)$ mapping to $1 \in \mathcal{O}_X(U_i)$. Then e_i defines a map $\mathcal{O}_Y|_{U_i} \to \mathcal{E}|_{U_i}$ whose kernel contains $\mathcal{I}^2$. In this way we see that $\mathcal{E}|_{U_i}$ comes from a pushout as above. This shows that c is surjective. We omit the proof of injectivity. $\square$